

JGR Earth Surface

RESEARCH ARTICLE

10.1029/2021JF006182

Key Points:

- The impact of a warmer, drier, or warmer and drier climate on streamflow and sediment flux was modeled for catchments in the high Andes
- Declines in streamflow and sediment flux exceeded predicted declines in precipitation and were largest in Argentina
- Impacts on streamflow and sediment flux were highly variable under a warming-only scenario, especially in Chile

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

J. R. Slosson,
jrslosson@syr.edu

Citation:

Slosson, J. R., Kelleher, C., & Hoke, G. D. (2021). Contrasting impacts of a Hotter and drier future on streamflow and catchment scale sediment flux in the high Andes. *Journal of Geophysical Research: Earth Surface*, 126, e2021JF006182. <https://doi.org/10.1029/2021JF006182>

Received 25 MAR 2021

Accepted 2 AUG 2021

Contrasting Impacts of a Hotter and Drier Future on Streamflow and Catchment Scale Sediment Flux in the High Andes

J. R. Slosson¹ , C. Kelleher¹ , and G. D. Hoke¹ 

¹Department of Earth and Environmental Sciences, Syracuse University, Syracuse, NY, USA

Abstract Fluvial sediment transport in high mountain regions is uniquely sensitive to changes in hydroclimate that alter the timing and magnitude of streamflow contributions from snowmelt, glaciers, and other cryoforms. In the high Andes of Argentina and Chile, mountain river systems serve millions of people in foothill communities and large metropolitan areas alike, yet these same rivers can also carry high sediment loads that compromise water quality and infrastructure. Regional climate predictions are also increasingly forecasting a warmer and drier future that is likely to alter hydrologic regimes and sediment transport dynamics. In this study, we use hydrologic modeling techniques and sediment rating curves compiled from historical data to propagate a suite of endmember climate projections to estimate changes in streamflow and consequent impacts on catchment-scale sediment transport on both the Argentine and Chilean sides of the high Andes. Our results show reductions in streamflow of up to 57% under scenarios with reduced precipitation and even larger decreases in sediment flux, while outcomes under a warming scenario with no changes to precipitation were highly variable. Our results further show that high winter streamflow events may increase in magnitude in Chile while mountain communities in Argentina may increasingly face water shortages. These results indicate that different adaptation and mitigation strategies may be needed on either side of the Andes in response to a changing hydroclimate.

Plain Language Summary Mountain streams in the high Andes play important roles sustaining foothill communities in Argentina and Chile by providing freshwater and supporting agriculture and power generation. However, these same streams can also carry high amounts of sediment, potentially harming water quality and leading to the infill of reservoirs that store water for use during dry times of the year. Future climate predictions in the region forecast increases in temperature and decreases in precipitation. These changes can have profound impacts on streamflow, which is heavily dependent on snow and glacier melt from the high mountains. How these shifts will influence sediment loads carried by mountain streams in the Andes is largely unknown. In this study, we model how expected changes in precipitation and temperature will alter streamflow and its effects on sediment transport. Models based on declines in precipitation show even larger declines in streamflow and sediment transport on both the Argentine and Chilean sides of the Andes. However, models based solely on temperature increases show highly variable outcomes, especially in Chile, where most of the yearly precipitation occurs during the winter. These results suggest that different planning strategies may be needed on the eastern and western sides of the Andes.

1. Introduction

Warming temperatures and future changes in precipitation across high mountain regions will have cascading effects through fluvial networks affecting the timing and magnitude of sediment exported from high relief catchments. These regions are uniquely sensitive to hydroclimatological changes due to the important contributions of glaciers, snow, and other cryoforms to fluvial sediment transport networks and the regulation of these processes by air temperature (East & Sankey, 2020; Hirschberg et al., 2021; Wulf et al., 2012). While efforts to quantify the impacts of climate change on sediment fluxes from mountainous and higher altitude regions have increased in recent years (e.g., Hirschberg et al., 2021; D. Li et al., 2020, 2021; Wulf et al., 2012), efforts to understand these processes in the high Andes Mountains of Argentina and Chile are still relatively sparse (e.g., Carrillo & Mao, 2020).

High relief, tectonically active mountainous regions like the Andes, where non-volcanic peaks reach altitudes of over 6,000 m, can generate significant quantities of sediment. Although suspended sediments from these areas help deliver important mineral nutrients to downstream ecological communities (McClain & Naiman, 2008), excess sediment loads can have negative impacts on aquatic environments. High sediment loads can impair aquatic biota (Bilotta & Brazier, 2008) and transport high concentrations of heavy metals (Dawson & Macklin, 1998), a particular concern for the copper-rich Andes. Excess sediment loads also pose problems for human consumption and infrastructure, requiring costly removal from drinking water sources (Bilotta & Brazier, 2008) and leading to infilling of reservoirs (Snyder et al., 2004). These impacts potentially pose threats to existing and planned critical infrastructure in and adjacent to the high Andes. Several large dams have been built in the region due to limited freshwater resources and feasibility studies for additional dams are already underway.

Sediment export in the high Andes is strongly seasonal and closely connected with fluvial processes (Carrillo & Mao, 2020; Mao & Carrillo, 2017; Pepin et al., 2010). Today, snowmelt is the primary source of streamflow in the region, although high elevation glaciers buffer streamflow during warmer and drier times of the year (Ayala et al., 2020; Carrillo & Mao, 2020; Crespo et al., 2020; Masiokas et al., 2006). Streamflow from the high Andes also supports a thriving agricultural sector, hydropower generation, and millions of people living in foothill communities, including the major metropolitan areas of Mendoza and Santiago (Crespo et al., 2020; Masiokas et al., 2006). As in other mountainous areas, because the timing and amount of annual runoff in the high Andes depends heavily on snowfall and glacier melt, it is sensitive to a changing climate (Cortés et al., 2011; Ragettli et al., 2016; Viviroli et al., 2011). In the future, climate change-induced shifts in regional hydrology, including modification of the proportion of precipitation events occurring as rain versus snow and loss of seasonal snow cover at lower elevations, will alter the dynamics of sediment delivery to stream channels (Iida et al., 2012). Quantifying these hydrologic shifts can be a key to understanding future changes in sediment transport and help anticipate mitigation needs, such as infrastructure changes or measures to preserve water quality.

Researchers have already observed rising air temperatures as well as shrinking snowpack and glacier mass over the last several decades in the high Andes (Ayala et al., 2020; Dussaillant et al., 2019). These trends are anticipated to persist in future decades, with increases of 1.2–4.0°C in mean annual temperature and decreases of 3%–30% in mean annual precipitation predicted for catchments in the high Andes by end of the 21st century (Bozkurt et al., 2018; Vicuña et al., 2011). In addition, glaciers in the high Andes are estimated to have passed the point where further temperature increases will lead to increases in streamflow from glacier melt, with glaciers already receding at high elevations (Ragettli et al., 2016). Under a hypothetical constant climate scenario, glacier mass reductions of nearly 20% are predicted for some parts of the high Andes by the end of the 21st century (Ayala et al., 2020), although these numbers climb to 70% under higher emissions scenarios (Ragettli et al., 2016). Likewise, annual runoff is predicted to decrease by 2%–3% under lower emissions scenarios and by as much as 40%–62% under higher emissions scenarios (Bozkurt et al., 2018; Ragettli et al., 2016). These changes are all likely to have concomitant impacts on water quality and sediment export.

Because the Andes act as an orographic barrier to precipitation originating from the Pacific (Masiokas et al., 2006), the western and eastern sides of the Andes have distinct climate regimes, potentially necessitating different responses to climate change. In this study, we use hydrologic models of eight catchments in the high Andes of Argentina and Chile to predict how streamflow will behave in an environment that is warmer, drier, or warmer and drier. We then use the outputs from those models to analyze how sediment transport through fluvial systems might change in comparison to a modern baseline period. The results offer insights into how broad shifts in precipitation or temperature trends may impact different parts of the Andes with contrasting hydroclimatologies.

2. Methods

2.1. Study Region

We analyzed eight catchments in the high Andes—four in Argentina and four in Chile—ranging from approximately 32°–35°S (Figure 1) based on the availability of paired streamflow and sediment flux data from

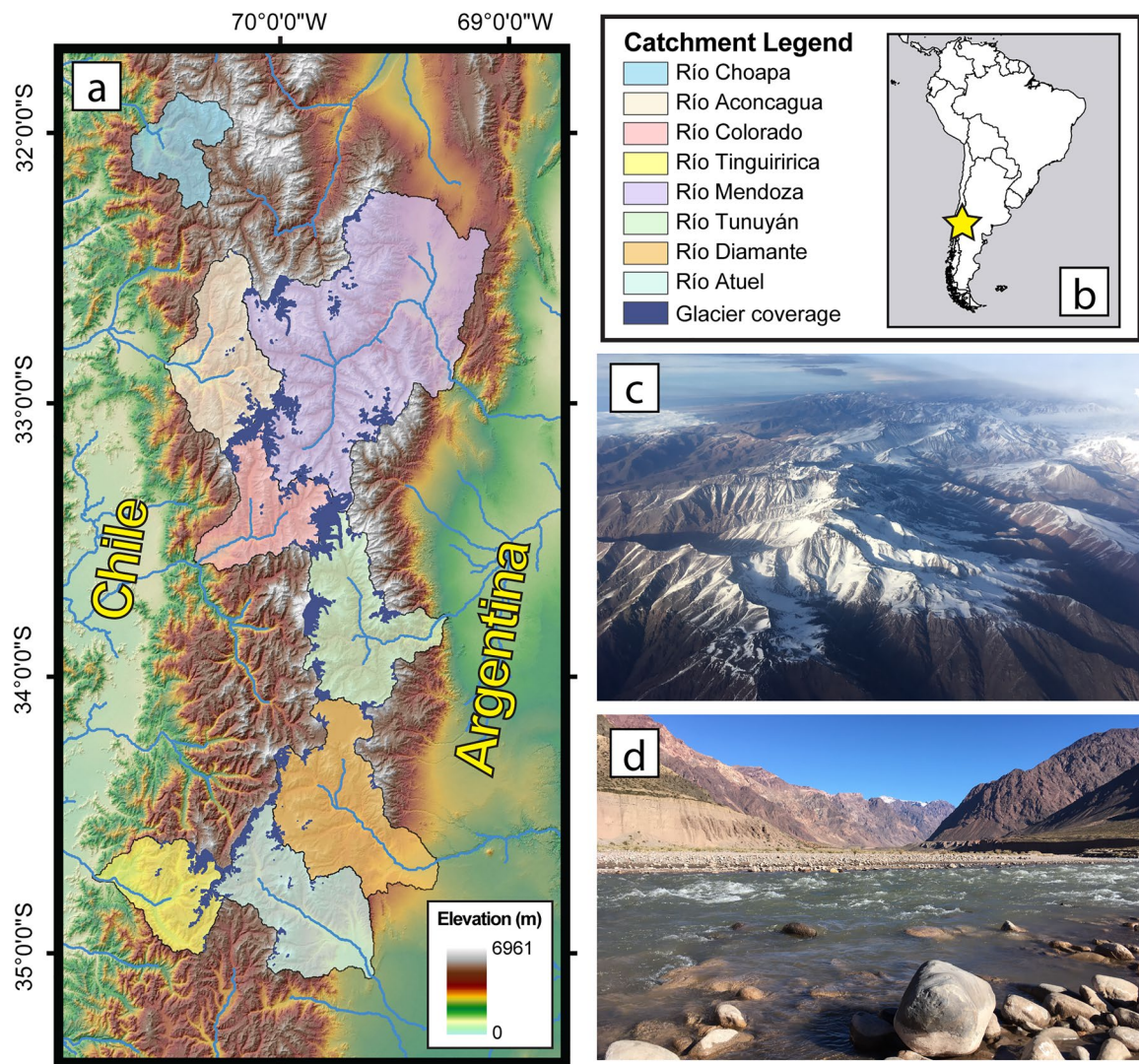


Figure 1. (a) Map of the study area in the high Andes of Chile and Argentina with shaded outlines of all eight catchments. The extent of glacier coverage within the catchments is shown in dark blue. Glacier coverage is from the Randolph Glacier Inventory (RGI Consortium, 2017). (b) Location of the study area within South America. (c) Aerial view of the study area looking south. (d) View of the Río Mendoza in the Andes Mountains between the towns of Uspallata and Polvaderas.

gauge station sites. No dams are located within the contributing areas of any of the catchments. We defined each catchment outlet using the location of its respective stream gauge station (Table 1) and delineated catchment boundaries using the D8 flow routing algorithm in ESRI ArcGIS 10.7 (ArcGIS) on the Shuttle Radar Topography Mission (STRM) 1 arc second digital elevation model (DEM). Gaps in the SRTM data were filled using the International Center for Tropical Agriculture 250 m DEM (Jarvis et al., 2008).

National agencies in both Argentina and Chile collect streamflow and sediment data. We obtained these data in Argentina from the Base de Datos Hidrológica Integrada (BDHI, 2020; SMRH, 2021), a publicly available resource maintained by the Secretaría de Infraestructura y Política Hídrica de la Nación. Streamflow data for the Chilean catchments are from the Catchment Attributes and MEteorology for Large-sample Studies data set in Chile (CAMELS_CL; Alvarez-Garretón et al., 2018), which includes a compilation of data from stream gauges maintained by the Dirección General de Aguas de Chile (DGA). We obtained suspended sediment load data for Chile from the DGA (DGA, 2020).

Table 1
Stream Gauge Stations and Associated Watershed Characteristics for All Eight Study Catchments

Gauge station	Gauge station number	Lat (°S)	Lon (°W)	Gauge station elevation (m)	Catchment mean elevation (m)	Catchment max elevation (m)	Catchment area (km ²)	Glacier cover (%)	Mean annual precip (mm/a)	Mean annual temp (°C)
Chile										
Río Choapa at Cuncumen	4703002	31.9667	70.5944	1,204	3,142	5,038	1,132	0.0	503	8.73
Río Aconcagua at Chacabuquito	5410002	32.8503	70.5094	939	3,178	5,924	2,113	2.26	506	8.55
Río Colorado prior to confluence with Río Maipo	5707002	33.5875	70.3669	962	3,419	6,550	1,663	9.12	473	6.96
Río Tinguiririca at Bajo Los Briones	6028001	34.7186	70.8267	573	2,550	4,962	1,438	5.08	1,090	7.45
Argentina										
Río Mendoza—at Guido	1413	32.9153	69.2378	1,419	3,528	6,961	7,155	4.24	240	3.23
Río Tunuyán—at Uco Valley	1419	33.7765	69.2725	1,225	3,649	6,532	2,458	9.83	332	0.81
Río Diamante—at La Jaula	1423	34.6685	69.3163	1,468	3,094	5,265	2,832	2.33	434	2.70
Río Atuel—at El Sosneado	1428	35.0839	69.6031	1,602	2,948	5,176	2,364	3.01	659	3.72

Note. Latitude and longitude are shown for the location of the gauge station at each catchment outlet. Total catchment glacier area from the Randolph Glacier Inventory (RGI Consortium, 2017). Mean annual precipitation for the water year 2005–2014 baseline period, averaged across each catchment, from the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) data set (Funk et al., 2015). Mean annual temperature for the Chilean catchments from the CR2MET database via the Catchment Attributes and Meteorology for Large-sample Studies data set in Chile (CAMELS_CL) database (Alvarez-Garretón et al., 2018). Mean annual temperature for the Argentine catchments from individual gauge stations (BDHI, 2020) and adjusted to the mean catchment elevation using a lapse rate of 0.6°C per 100 m.

The eight study catchments range in size from ~1,000 to 7,000 km² and encompass elevations ranging from 573 to 6,961 m (Table 1). The catchments generally have either sparse or no vegetation, although there are small regions of shrubland or herbaceous vegetation near some trunk streams and at lower elevations (Buchhorn et al., 2020). Glaciers are present at higher elevations, but do not exceed 10% of the total area of any single watershed (Table 1; RGI Consortium, 2017). Across the study region, precipitation broadly increases from north to south (Bozkurt et al., 2018) and from lower to higher elevations (Vicuña et al., 2011), with the western side of the Andes receiving more annual precipitation than the eastern side (Masiokas et al., 2006). Precipitation is winter-dominated in central Chile and increases from 290 mm annually in Santiago at 33°S in the Central Valley (Masiokas et al., 2006) to over 1,500 mm annually between 35° and 36°S on the western slopes of the Andes (Bozkurt et al., 2018). Winter storms in Chile can be severe, sometimes delivering over 100 mm of precipitation in a single event (Viale & Norte, 2009). In Argentina, precipitation during the winter months spills over from the west at higher elevations. A different pattern prevails during the summer in the eastern Andes, with precipitation increasing at lower elevations due to convective storms and moist air masses originating from the northeast (Masiokas et al., 2006; Viale & Nuñez, 2011). Annual precipitation also exhibits significant interannual variability on either side of the range, with higher annual precipitation associated with El Niño events and lower annual precipitation associated with La Niña (Bozkurt et al., 2018; Cortés et al., 2011; Masiokas et al., 2006). Previous estimates of catchment-averaged erosion rates in the high Andes approach 0.30 mm/a and are identical within uncertainty using either ¹⁰Be or stream gauge data (Carretier et al., 2013; Val et al., 2018). Erosion rates reflect tectonic processes within the region and are highest between 33° and 34°S (Val et al., 2018).

2.2. Hydrologic Models

We generated simulations and predictions of streamflow using the Hydrologiska Byråns Vattenavdelning (HBV) catchment-runoff model (Bergström, 1992; Seibert & Vis, 2012). While there are many versions of HBV, we implemented a semi-distributed framework with 15 parameters that requires daily precipitation, temperature, and potential evapotranspiration (PET) as inputs and produces streamflow as output for a given catchment. To account for the impact of elevation on snow accumulation and melt, the modeled catchments were divided into a series of 500 m vertical elevation bands in ArcGIS. Within each band,

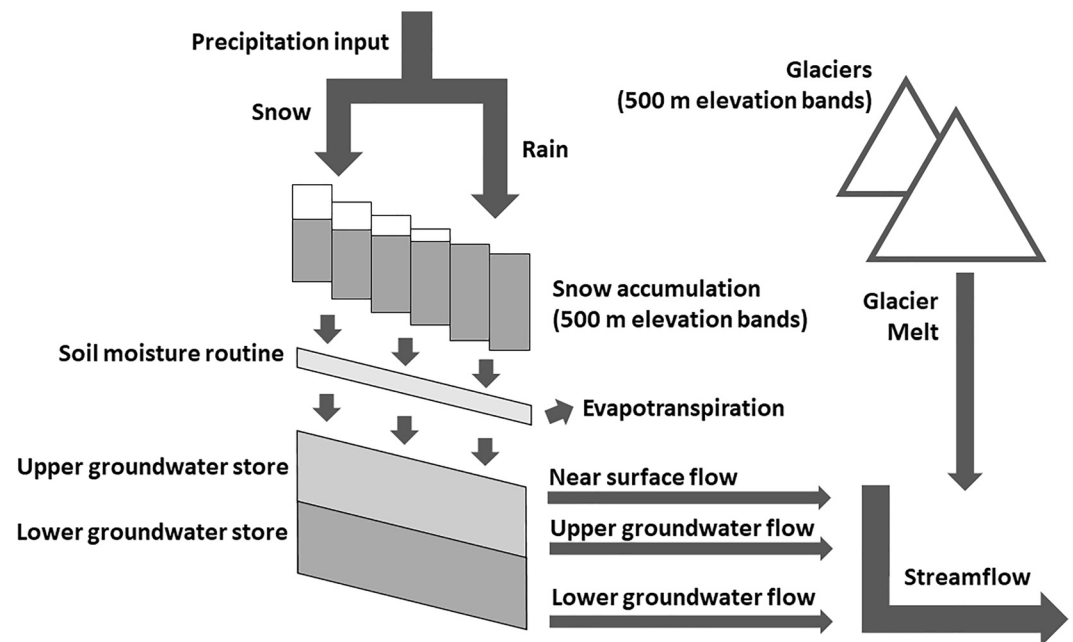


Figure 2. Simplified conceptual overview of the Hydrologiska Byråns Vattenavdelning (HBV) model implementation used in this study, based on Seibert and Vis (2012).

precipitation falls as snow at elevations where the observed temperature is below a specified threshold temperature (model parameter), typically close to 0°C. Precipitation stored as snow melts at a rate determined by the degree-day factor (model parameter) when the temperature exceeds this threshold. Melt is then treated (like rainfall) as an input to a soil moisture store, which is routed to subsurface stores before finally being routed to streamflow. Figure 2 shows a conceptual diagram of the HBV model used in this study. Additional information on model mechanics is available in the Supporting Information.

We obtained climate data for the Argentine and Chilean catchments using similar methods, where practicable. For Chile, all climate data and a variety of watershed attributes were obtained via CAMELS_CL (Alvarez-Garreton et al., 2018). Precipitation data for the Argentine catchments were compiled using the Climate Hazards Group InfraRed Precipitation with Stations database (CHIRPS), which provides daily precipitation estimates at a 0.05° resolution (Funk et al., 2015) and has been validated by other studies in the high Andes (Rivera et al., 2018). These data were clipped to the Argentine catchments and averaged to create a single daily value per catchment. CHIRPS data for the Chilean catchments were obtained via CAMELS_CL. For Argentina, we compiled air temperature data from individual gauging stations within or in close proximity to each catchment. The national CR2MET data set via CAMELS_CL was used for daily air temperature data for Chile. We calculated monthly PET values for the Argentine catchments using the Thornthwaite method (Thornthwaite & Mather, 1955) applied to gauge station air temperature observations. For the Chilean catchments, we obtained monthly PET from a temperature-based compilation via CAMELS_CL. Temperature inputs to the HBV models were scaled vertically across each catchment elevation band using a lapse rate of 0.6°C per 100 m, similar to observed average lapse rates in the region (Ibañez et al., 2020). A precipitation gradient of 10% per 100 m was used to reflect orographic effects in the Andes (Viale & Garreaud, 2015).

Because all of the catchments except the Río Choapa contain glaciers, we coupled a simple glacier model to our HBV framework to account for the glacier contribution to streamflow. For the Argentine catchments, the extent of glacial coverage within each catchment was calculated using the Randolph Glacier Inventory (v. 6.0; RGI Consortium, 2017). For the Chilean catchments, we obtained these data from CAMELS_CL. Glacier area was divided into elevation bands using the same methods as for the HBV routine. In the model, glaciers produce meltwater whenever the temperature rises above a temperature threshold (equivalent to the melt threshold for snow) at a rate equal to a calibrated melt coefficient multiplied by the glacier area.

Glacier melt is routed directly to streamflow. In models of future streamflow coupled with reductions in the glacier area, we reduced the total glacier area by a specified percentage based on assumptions for a given climate outcome. This reduction was achieved by removing glacier coverage from the lowest elevation band containing glacier mass in each catchment until the total glacier area in that elevation band was removed. This process repeated through progressively higher elevation bands until reaching the total specified percentage reduction.

2.3. Model Calibration

We calibrated models to observed streamflow for each catchment using a baseline period of April 1, 2005–March 31, 2015. The period of April 1 through March 31 corresponds to the regional water year (Alvarez-Garretón et al., 2018; Cortés et al., 2011). This baseline period includes the 2010–2015 “mega drought” in the high Andes (Garreaud et al., 2017; Rivera et al., 2017), which enabled model calibration to both wet and dry years. Daily observed streamflow records were at least 85% complete for all catchments during this interval. An initial set of model parameters for each catchment were optimized using a Genetic Algorithm and Powell (GAP) routine, as described in the Supporting Information. To account for potential equifinality (i.e., different combinations of parameters generating equivalent performance metrics), we then generated 2,000 parameter sets using Latin hypercube sampling from ranges of $\pm 30\%$ applied to optimized parameter values (to the extent allowable within individual parameter limits in HBV) and corresponding simulations for the baseline period. Within each catchment, we identified the top 100 parameter sets corresponding to the highest Nash-Sutcliffe efficiency (NSE; Nash & Sutcliffe, 1970). The top 100 values for the glacier melt coefficient were determined similarly, based on Latin hypercube sampling of 1,000 values evaluated and ranked based on NSE from an initial parameter space determined through trial and error. We used these top 100 parameter sets as the basis for subsequent scenario analysis. Instead of using a single, optimized parameter set, multiple parameter sets were included to account for parameter uncertainty as well as uncertainty introduced when modeling future periods based on historical performance. NSE was used as the primary error metric based on visual observation of model performance and the strong seasonality of the annual hydrographs. Kling-Gupta efficiency (KGE) was computed as a secondary error metric (Gupta et al., 2009). We assumed the selected parameter sets captured historical variability and used them to predict streamflow response to future climate scenarios (Singh et al., 2014). Additional details on the calibration routine and error metrics are included in the Supporting Information.

2.4. Scenario Analysis

We used three end-member climate scenarios to model changes in streamflow and sediment flux based on temperature and precipitation forecasts for the region reported in prior studies (Bozkurt et al., 2018; Vicuña et al., 2011). Although we do not assign exact dates to these climate outcomes, the precipitation and temperature forecasts are keyed to expectations under higher emissions scenarios by the end of the 21st century. These three scenarios were chosen due to uncertainty as to whether changes in precipitation, temperature, or both will dominate. Our drier scenario models a 30% reduction in mean annual precipitation but no change in mean annual temperature. This scenario includes a 20% reduction in total glacier area based on prior work in the region modeling glacier retreat under steady-state assumptions (Ayala et al., 2020). Our drier and warmer scenario models a 30% reduction in mean annual precipitation coupled with 4°C mean annual temperature increase. Finally, our warmer scenario features a 4°C mean annual temperature increase but no change in mean annual precipitation. In both the drier and warmer scenario and the warmer scenario, the total glacier area is reduced by 70%, an assumption in line with projections for the region based on higher IPCC emissions scenarios (Ragettli et al., 2016).

For each scenario and each catchment, we created 50 time series of daily weather data using the daily stochastic weather generator WeaGETS, which also simulates interannual variability (Chen et al., 2012). Observed daily weather data for the baseline period were used as input to the weather generator, uniformly modified by the increases in temperature or decreases in precipitation noted above, both for simplicity and based on prior modeling work showing that decreases in precipitation occurred in each month of the year under higher emissions scenarios (Bozkurt et al., 2018). The top 100 parameter sets were then combined

with the 50 climate time series generated for each scenario, resulting in 5,000 simulations of streamflow for each climate scenario per catchment.

2.5. Hydrologic Signatures and Erosion Rates

To obtain a holistic view of how changes to precipitation and temperature alter streamflow dynamics in each watershed, we calculated several hydrologic signatures for the baseline period and for each future modeled time series of streamflow. These signatures include:

1. The highest observed daily discharge for each time series, as these events can be particularly important for transporting sediment (Leopold et al., 1964; Molnar et al., 2006).
2. The center of volume (Court, 1962), to investigate how warming temperatures affect streamflow timing due to snow and glacier melt. Center of volume was calculated as the day, beginning with the April 1 water year, on which half of the total streamflow for a given water year passes the catchment gauge station (Court, 1962).
3. Unit stream power (Bagnold, 1966), a metric of each stream's ability to perform geomorphic work, using the following formulation:

$$\omega = \frac{\Omega}{\text{channel width}} = \frac{\rho g Q S}{\text{channel width}} \quad (1)$$

where ω is unit stream power (W/m^2), Ω is total stream power (W/m), ρ is the density of water ($1,000 \text{ kg/m}^3$), g is gravitational acceleration (9.8 m/s^2), Q is streamflow (m^3/s), and S is the slope of the streambed (m/m ; Bagnold, 1966). Streambed slope was calculated in ArcGIS by measuring the change in elevation along a $\sim 500 \text{ m}$ long stream channel segment centered on each gauge station location and channel width was assessed using satellite imagery.

4. The average position of the 0°C isotherm during the winter months, to interpret how warming temperatures affect the distribution of rain versus snow events. Observed temperature data from the baseline period and simulated temperature data for the warming scenarios were used in these calculations. Following other regional studies, we define the winter season as June through September (JJAS; Rusticucci et al., 2014).
5. The relative change in catchment areas exposed to temperatures above freezing, on average, from the baseline period to the warming scenarios, calculated by comparing the change in the location of the JJAS 0°C isotherm in each catchment to the respective catchment hypsometry derived from ArcGIS.
6. The average slope of each catchment below the baseline period JJAS 0°C isotherm and for the area between the baseline period JJAS 0°C isotherm and warming scenario JJAS 0°C isotherms, to investigate areas of potential slope instability.
7. Erosion rates, calculated using observed and simulated hydrographs and sediment rating curves developed for each catchment (Gao, 2008; L. Li et al., 2020; Runkel et al., 2004), as a normalized metric of sediment flux and for ease of comparison with prior studies. We applied a power-law fit to paired river discharge and sediment load data per catchment and then applied that fit to streamflow for the baseline period and model outputs, using the following formulation:

$$Q_s = aQ^b \quad (2)$$

where Q_s is sediment flux, Q is the streamflow, and a and b are rating curve parameters (Gao, 2008). Other researchers in the region have followed this approach using similar data sets (Val et al., 2018). We corrected for rating curve bias, as appropriate, using methods described in the Supporting Information (Ferguson, 1986; Runkel et al., 2004).

3. Results

3.1. Water Year 2005–2014 Climate Variables and Streamflow

Precipitation, temperature, and streamflow data for the baseline period are representative of the contrasting climate characteristics across the study region. Monthly averaged precipitation was higher and peaked during the winter months in Chile, while average precipitation was lower in Argentina (Figure 3 and Table 1).

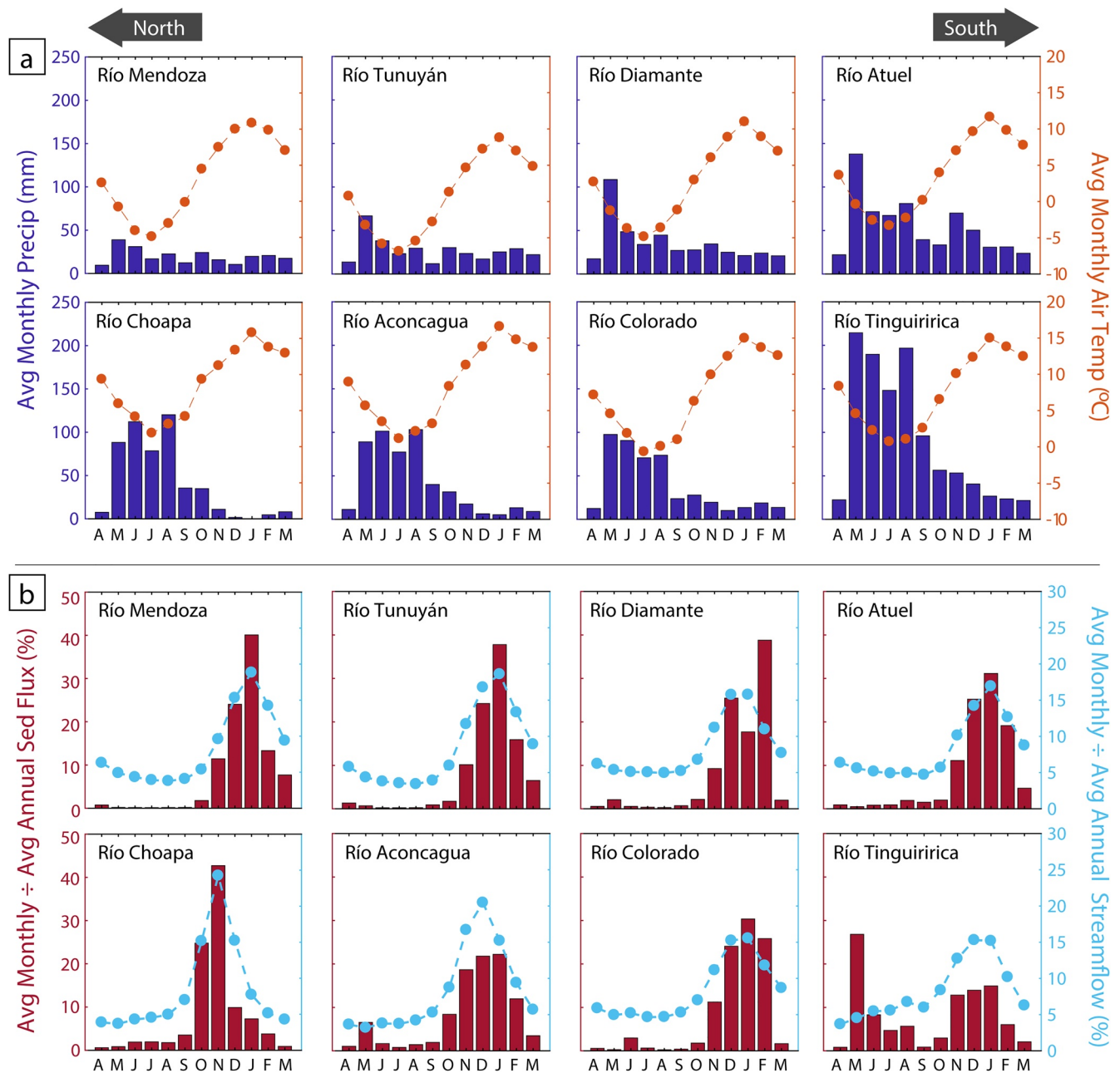


Figure 3. (a) Average monthly baseline period (water year 2005–2014) precipitation and temperature data for each watershed. (b) Average monthly baseline period (water year 2005–2014) streamflow and sediment flux data for each watershed, shown as a percentage of total average annual streamflow and sediment flux, respectively, for each watershed for ease of comparison. Note that sediment flux data for the Río Colorado used in this study are only available through 2010 (Table S1). For (a) and (b), Argentine catchments appear on the first row, followed by Chilean catchments on the second row.

On both sides of the Andes, precipitation generally increased heading south. Following the winter season, average monthly streamflow increased during the austral spring in association with snowmelt and peaked during the summer, although peak flows arrived earlier in the unglaciated Río Choapa catchment. Peaks in average monthly sediment flux were strongly correlated with streamflow and, with the exception of the Río Tinguiririca, very little sediment was transported in the fluvial system from April through October (Figure 3).

Hydrographs for all eight catchments reflect the snowmelt-dominated regime of each basin (Figure 4). On both sides of the range, the catchments with the highest mean elevations also generally had the greatest

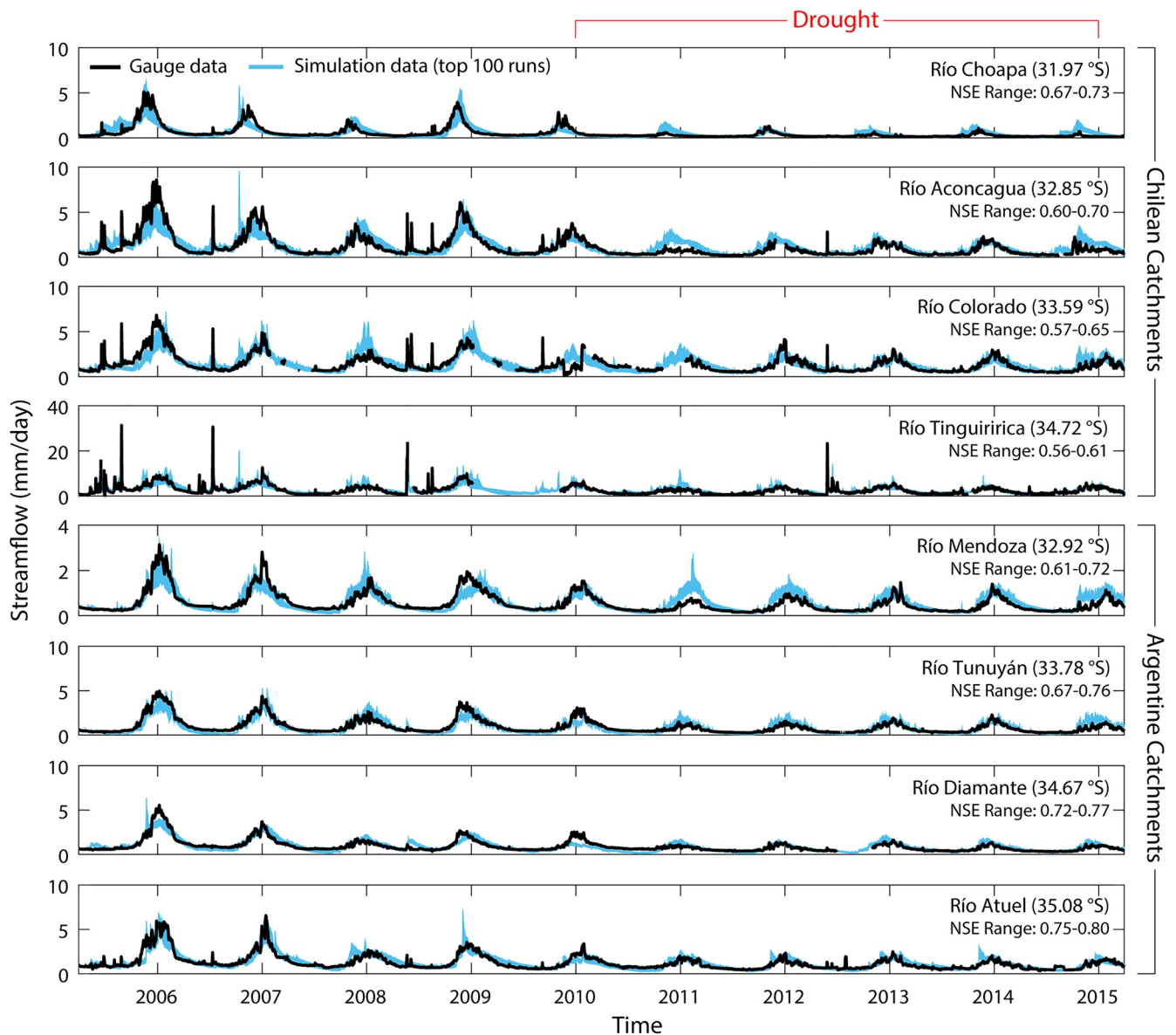


Figure 4. Streamflow (mm/day) from gauge data and all top 100 simulations for water year 2005–2014 baseline period. X-axis ticks denote start of each listed calendar year. Note differences in y-axis scale among catchments and reductions in peak flow during the 2010–2015 “mega drought.” Latitude of each catchment listed parenthetically. Nash-Sutcliffe Efficiency (NSE) ranges for the top 100 parameter sets are listed for each catchment.

total glacier contribution to streamflow by volume (Tables 1 and 2). Total annual streamflow showed an observable decline in each catchment starting around the water year 2010, coincident with the regional “mega-drought” spanning from 2010 to 2015.

In Chile, there were several large, singular discharge events that are distinct from the sustained snowmelt portion of the annual hydrograph. These events were pronounced in the Río Colorado, Río Aconcagua, and Río Tinguiririca catchments and appear to occur nearly simultaneously in all three catchments. In the case of the Río Tinguiririca, discharge associated with these events in some cases exceeded the maximum daily discharge associated with the snowmelt portion of the hydrograph. One such event on August 27, 2005 is coincident with a large and well-documented winter storm that caused landsliding at lower elevations, flooding in Santiago, and led to the evacuation of city residents (Viale & Norte, 2009). Such singularly large discharge events are not as evident in the Argentine hydrographs.

Table 2

Mean Annual Total Runoff (Q) for Each Catchment for Observed and Top 100 Simulated Baseline Period Time Series and All Future Climate Scenarios

Catchment	Mean annual observed runoff (mm/a)	Mean annual model runoff and min, max (mm/a)	Mean annual glacier contribution and min, max (mm/a)	Mean annual glacier contribution (m ³ /a)	Model KGE range	Model NSE range	Drier scenario mean annual Q (mm/a) and % change	Drier & warmer scenario mean annual Q (mm/a) and % change	Warmer scenario mean annual Q (mm/a) and % change
Chile									
Choapa	196	178 (137, 228)	0	0	0.56, 0.84	0.67, 0.73	102 (−43%)	86 (−52%)	202 (+13%)
Aconcagua	410	411 (358, 458)	72 (67, 77)	152 × 10 ⁶	0.58, 0.75	0.60, 0.70	268 (−35%)	222 (−46%)	364 (−11%)
Colorado	492	488 (442, 528)	39 (26, 52)	65 × 10 ⁶	0.64, 0.80	0.57, 0.65	329 (−33%)	289 (−41%)	442 (−9%)
Tinguiririca	952	954 (852, 1,064)	37 (30, 44)	53 × 10 ⁶	0.60, 0.76	0.56, 0.61	645 (−33%)	612 (−36%)	937 (−2%)
Argentina									
Mendoza	201	205 (181, 223)	20 (4, 36)	140 × 10 ⁶	0.65, 0.84	0.61, 0.72	121 (−41%)	91 (−56%)	149 (−27%)
Tunuyán	340	297 (276, 316)	43 (33, 54)	106 × 10 ⁶	0.74, 0.85	0.67, 0.76	174 (−41%)	127 (−57%)	201 (−32%)
Diamante	336	332 (296, 349)	21 (16, 26)	60 × 10 ⁶	0.76, 0.87	0.72, 0.77	189 (−41%)	152 (−53%)	256 (−20%)
Atuel	451	450 (404, 503)	11 (5, 16)	25 × 10 ⁶	0.75, 0.89	0.75, 0.80	250 (−44%)	203 (−55%)	362 (−19%)

Note. Drier scenario = −30% precipitation. Drier and warmer scenario = −30% precipitation +4°C. Warmer scenario = +4°C. Minimum and maximum Kling-Gupta Efficiency (KGE) and Nash-Sutcliffe Efficiency (NSE) values for top 100 models are also shown for each catchment. Percent changes for future climate scenarios are based on a comparison of mean simulated values for the baseline period to the mean of the future scenario. Both area-normalized and total volume of glacier contribution to baseline period streamflow are also shown.

Visually, the models of streamflow for the baseline period from the top 100 parameter sets are similar to observed annual patterns of streamflow for the different catchments (Figure 4). Model performance is summarized for all catchments in Table 2. Observed mean annual streamflow and maximum daily streamflow values for the baseline period were within the range of modeled values for nearly all catchments (Tables 2 and S5). The highest NSE values for the top 100 simulations ranged from 0.61 (Río Tinguiririca) to 0.80 (Río Atuel); the highest performing KGE values ranged from 0.75 (Río Aconcagua) to 0.89 (Río Atuel; Table 2).

3.2. Future Climate Scenario Streamflow Simulations

Future simulations broadly show reductions in streamflow for all scenarios (Table 2). Reductions are highest for the drier and warmer scenario. Average annual streamflow reductions under this scenario range from −36% to −57%, with higher reductions observed on the drier Argentine side of the range. Reductions in streamflow under the drier scenario are also large, ranging from −33% to −44%. The reductions in streamflow under the drier scenario and drier and warmer scenario exceed the 30% reduction in precipitation used in these scenarios.

Reductions in annual streamflow are considerably more variable for the warmer scenario. On the drier Argentine side of the range, predicted streamflow reductions range from −19% to −32% for the four catchments. In contrast, streamflow reductions on the wetter, winter precipitation driven Chilean side are modest, ranging from −11% for the Río Aconcagua to −2% for the Río Tinguiririca. In the Río Choapa catchment, which does not contain glaciers and therefore did not lose this important water supply, the average annual streamflow increases under the warmer scenario by 13%. Changes in streamflow in the future scenarios are mirrored by reductions in unit stream power for nearly all of the catchments, with a modest increase in unit stream power for the Río Choapa under the warmer scenario (Figure 5a and Table S6).

Maximum daily stream discharge decreases in the drier scenario and drier and warmer scenario for all catchments except the Río Tinguiririca (Figure 5b and Table S5). These decreases are more modest in the wetter Chilean catchments as compared to the Argentine catchments. In contrast, there is more variability in maximum discharge events under the warmer scenario. For the Chilean catchments, these events are

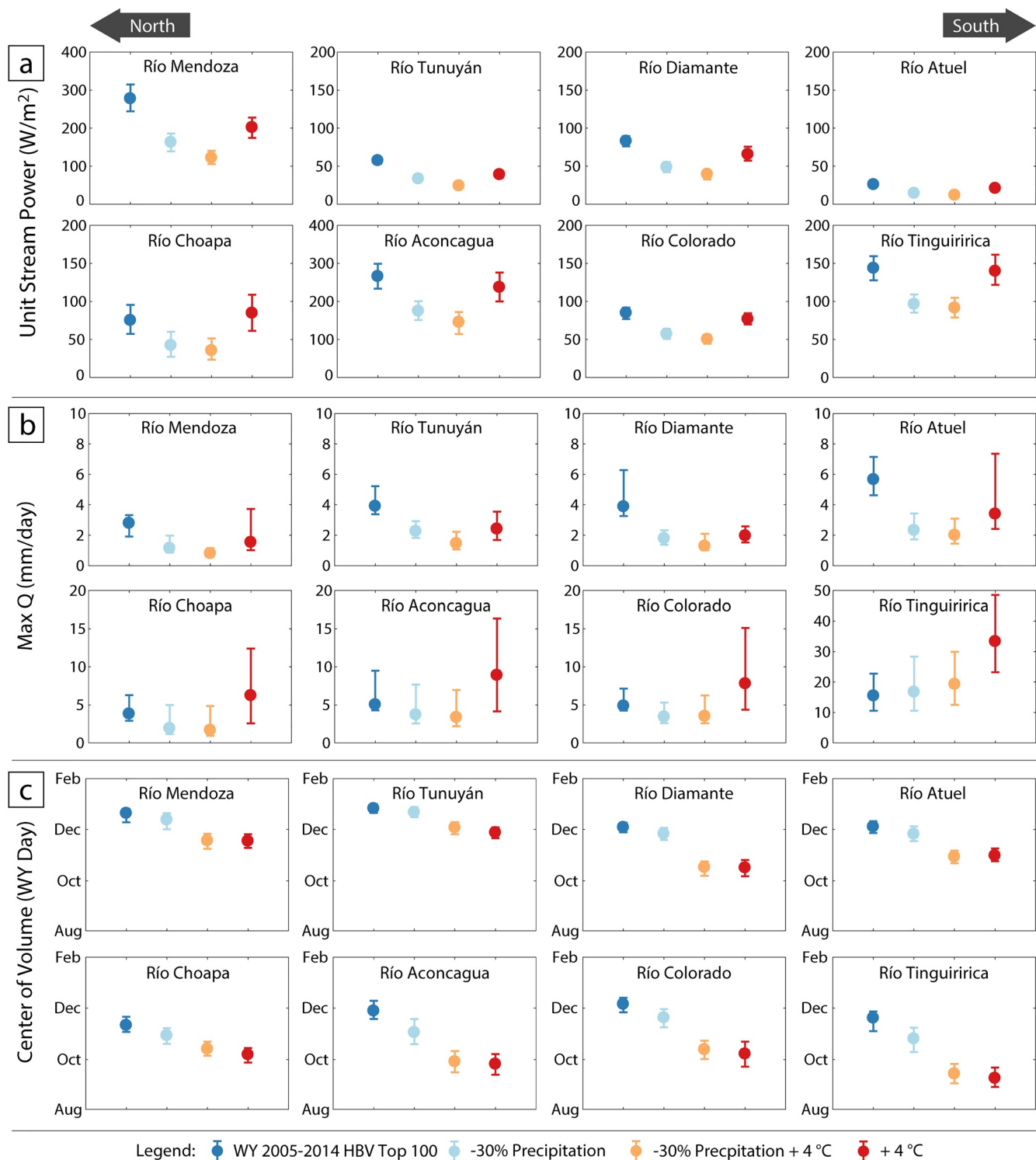


Figure 5. (a) Unit stream power, (b) maximum stream discharge, and (c) center of volume results for top baseline period model runs ($n = 100$) and range of all results from 5th to 95th percentiles for future climate scenarios ($n = 4,500$). For each statistic, Argentine catchments appear on the first row, followed by Chilean catchments on the second row. In the results for maximum stream discharge events, note different y-axis limits for the Argentine and Chilean catchments and the even larger y-axis limit for the Río Tinguiririca as compared to all other catchments. Also, note different y-axis limits for the Río Mendoza and Río Aconcagua in the unit stream power results.

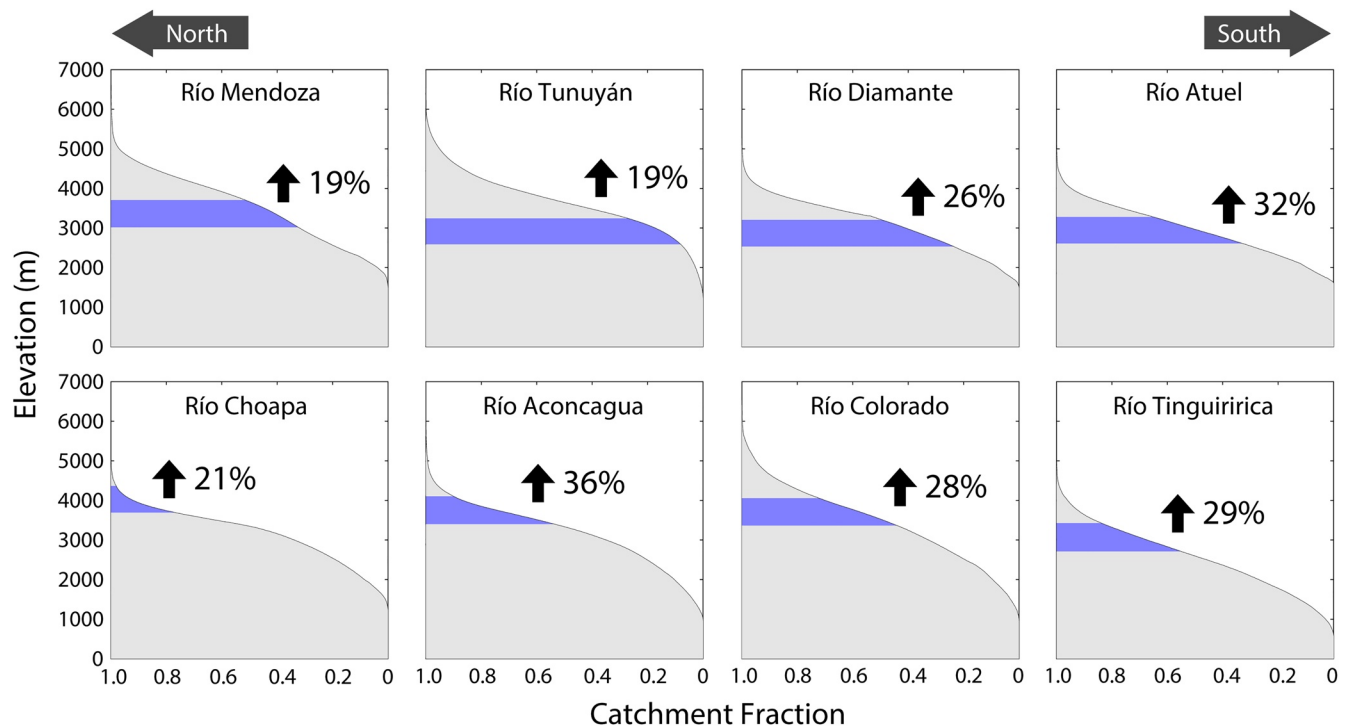


Figure 6. Catchment hypsometry and change in mean 0°C isotherm for each watershed for the June, July, August, and September (JJAS) time period from the baseline period to the $+4^{\circ}\text{C}$ warming scenarios. Argentine catchments appear on the first row, followed by Chilean catchments on the second row. Catchment hypsometry is shown in gray. The regions newly exposed to temperatures above freezing during this time period, on average, are shown in blue. The bottom of the blue shaded region represents the mean elevation of the baseline period JJAS 0°C isotherm and the top of the blue shaded region represents the mean elevation of the JJAS 0°C isotherm under the warming scenarios. The total percentage of the catchment area exposed to temperatures above freezing relative to the baseline period is also displayed for each catchment.

also typically larger in magnitude. For all of the Chilean catchments, the median from the range of maximum daily discharge events under the warmer scenario is higher than the baseline period. The very highest maximum daily discharge events from the range of warmer scenario simulations are also nearly two times as high as the warmer scenario medians. While the Argentine catchments also displayed some increased variability in maximum discharge under the warmer scenario, these changes are smaller compared to those in Chile. Indeed, the median from the range of maximum discharge events under the warmer scenario is lower for the Argentine catchments than the baseline period.

3.3. Center of Volume and 0°C Isotherm

As compared to baseline conditions, the center of volume timing (averaged across all simulations) is predicted to shift earlier in the year for all scenarios on both sides of the Andes (Figure 5c and Table S7). This shift is modest for a drier future, but more pronounced for both warming scenarios. For all of the Argentine catchments combined, we predict that the median of the average center of volume will arrive 1 week earlier for the drier scenario and 5 weeks earlier for both warming scenarios as compared to baseline. This earlier shift in the center of volume is more pronounced for the Chilean catchments. In these catchments, the center of volume is predicted to arrive approximately 3 weeks earlier for the drier scenario and 8 weeks earlier for both warming scenarios.

Model-predicted changes to the center of volume timing associated with both future warming scenarios are also predicted to be accompanied by a several hundred meters retreat in the JJAS 0°C isotherm for all catchments (Figure 6). Based on the hypsometry of each watershed, this exposes an average of 19%–36% of total catchment area to temperatures above freezing during the period when precipitation is more likely to fall as snow today; the magnitude of these changes are similar for both the Argentine and Chilean catchments. The topographic slope of this newly exposed area is also steeper for 75% of the catchments than the area

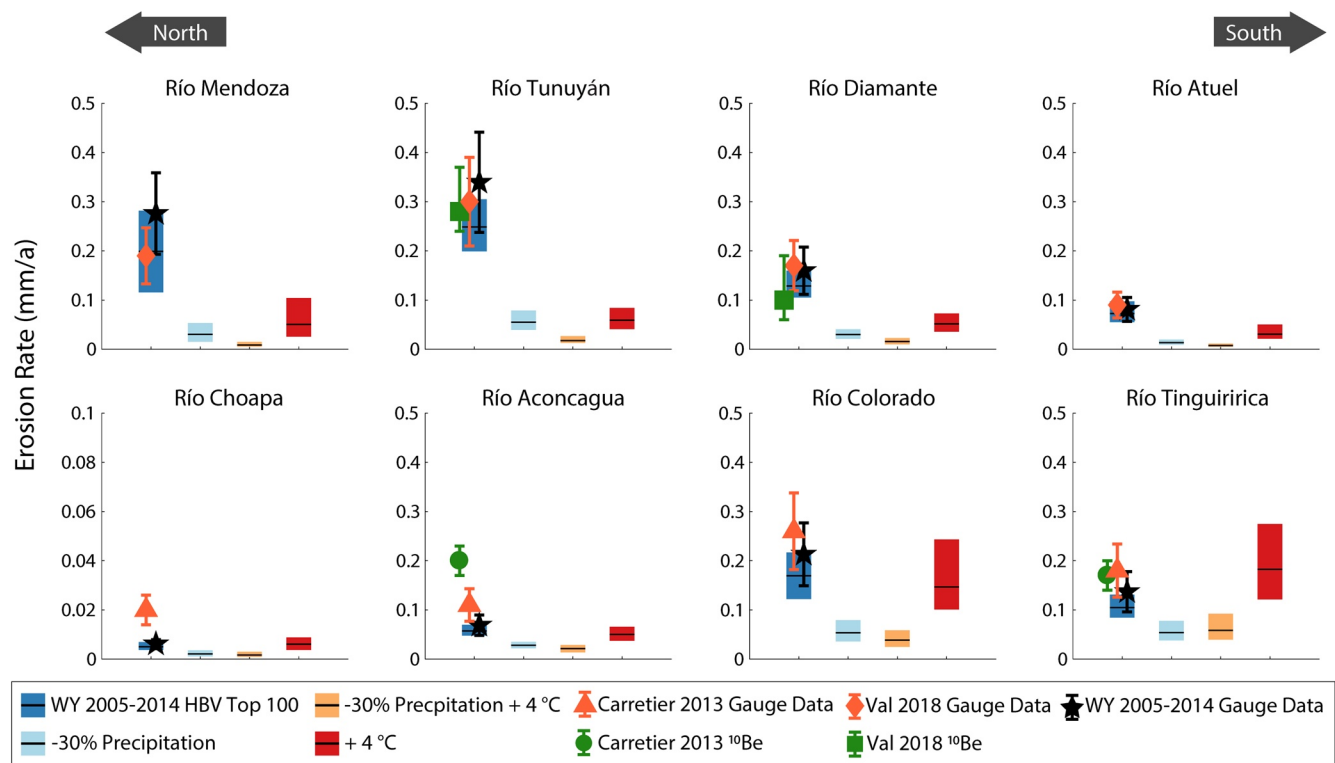


Figure 7. Simulated erosion rates for top baseline period model runs ($n = 100$) and range of all results from 5th to 95th percentiles for future climate scenarios ($n = 4,500$). Argentine catchments appear on the first row, followed by Chilean catchments on the second row. Black lines across vertical bars represent median erosion rates for each scenario depicted. Erosion rates calculated using observed stream gauge data from this and other studies along with ^{10}Be results from other studies (Carretier et al., 2013; Val et al., 2018) are shown for comparison to the baseline period scenarios. Note different y-axis limits for the Río Choapa, the northernmost catchment in this study, which also has the lowest observed erosion rates.

below the average baseline JJAS 0°C isotherm. For all catchments, the mean slope of the newly exposed area is $>20^{\circ}$; for five catchments, it is in excess of 25° (Table S9).

3.4. Sediment Fluxes and Erosion Rates

Baseline erosion rates calculated using observed streamflow are highest for the Río Colorado on the Chilean side of the range and the Río Tunuyán on the Argentine side of the range and decrease to the north and south (Figure 7 and Table S8). Median erosion rates calculated using the simulated streamflow time series for the baseline period are within error of the erosion rates calculated with observed streamflow data. The erosion rates calculated using modeled streamflow reproduce the trends calculated from observed streamflow from the baseline period.

Under future climate scenarios, reductions in streamflow and stream power in the drier scenario (-30% precipitation) and drier and warmer scenario (-30% precipitation + 4°C) propagate into large declines in sediment fluxes and erosion rates in every catchment (Figure 7 and Table S8). Declines are largest on the drier Argentine side of the Andes, ranging from 77% to over 90% as compared to baseline. Even on the Chilean side of the Andes, declines in erosion rates still range from 44% to 77%. The range of these predicted erosion rates is impressively narrow under these two scenarios.

The warmer scenario ($+4^{\circ}\text{C}$) produced variable responses with a greater range than both wetter future scenarios, reflecting increased variability in streamflow, stream power, and maximum discharge events. On the dry Argentine side of the range, reductions in erosion rates are still large, exceeding 70% for the high, glaciated Río Mendoza and Río Tunuyán catchments. On the wetter Chilean side, the Río Aconcagua and Río Colorado catchments were predicted to have modest decreases in erosion rates of $<15\%$ relative to the

baseline period. Erosion rates increased in the unglaciated Río Choapa catchment and also in the Río Tinguiririca catchment, where mean annual precipitation is highest.

4. Discussion

4.1. Changing Streamflow Regimes and Longitudinal Contrasts Across the High Andes

A variety of different hydrologic modeling approaches have been used to simulate streamflow in the high Andes, including the Water Evaluation and Planning (Vicuña et al., 2011), Variable Infiltration Capacity (Bozkurt et al., 2018), and the TOPKAPI-ETH version of the TOPographic Kinematic APPROXimation and Integration (TOPKAPI) models (Ayala et al., 2020; Ragettli et al., 2016). While these modeling frameworks all have varying degrees of complexity, an advantage of HBV is the relative ease of adaptation for use in modeling multiple catchments. The static nature of the assumptions used in this study mean that the framework is not suitable for modeling catchment behavior on a continuum through time, but rather as an investigation of potential endmember responses to future change. Use of weather generator-based precipitation and temperature inputs scaled by elevation also mean that some real-world differences in weather patterns across individual catchments, such as storm events that may only reach upper or lower elevations, are necessarily simplified.

Despite differences across modeling frameworks, our streamflow results show broad similarities with prior studies. Importantly, we also expand upon prior works by considering different climate outcomes and extending our analysis to additional catchments, including on the Argentine side of the Andes. In a study of four different basins on the Chilean side of the Andes stretching from 33° to 38°S, Bozkurt et al. (2018) predicted reductions in runoff of 40% by the end of the 21st century under the IPCC representative concentration pathway (RCP) 8.5 “business-as-usual” emissions scenario. Similarly, in a study of the Río Juncal catchment in Central Chile, a tributary of the Río Aconcagua, Ragettli et al. (2016) predicted that annual runoff would decrease between 40% (RCP 4.5) and 60% (RCP 8.5) by the end of the 21st century. For the Chilean catchments, our results suggest that annual streamflow will decline by 36%–52% under the drier and warmer scenario, most akin to RCP 8.5 “business-as-usual” assumptions (Table 2). The similarity of our predictions to prior works confirms that streamflow reductions in the Andes will be significant and exceed forecast reductions in precipitation (Bozkurt et al., 2018; Ragettli et al., 2016; Vicuña et al., 2011). Results from this study further show that declines in streamflow may be even larger on the Argentine side of the range (Table 2), indicating that different mitigation and adaptation strategies may need to be employed there than for catchments draining the western side of the Andes.

For the glaciated catchments, future anticipated streamflow reductions are exacerbated by shrinking glacier mass (Ragettli et al., 2016). In addition, enhanced evapotranspiration under the warming scenarios likely amplifies future streamflow response (Vicuña et al., 2011). Finally, with less annual precipitation, a greater fraction of incoming precipitation may simply replenish groundwater stores that are likely to become increasingly depleted (Thomas & Famiglietti, 2019). Surface water losses in the future may mean that the groundwater system will become an even more important source of freshwater in these high mountain catchments (Somers et al., 2019). While some efforts have been made to analyze the groundwater contribution to streamflow in the Argentine high Andes (e.g., Crespo et al., 2020), this may become an even more critical need in the coming decades.

Prior studies also predict an earlier arrival of peak runoff volume by weeks to months (Bozkurt et al., 2018; Ragettli et al., 2016), though our results show meaningful differences between the Chilean and Argentine catchments. While the average center of volume is predicted to arrive over a month earlier for the Argentine catchments under the drier and warmer scenario and warmer scenario, this shift was nearly 2 months earlier for the Chilean catchments (Figure 5 and Table S7). The earlier center of volume timing is linked with the upwards retreat of the mean elevation of the JJAS 0°C isotherm (Figure 6). With a higher 0°C isotherm, an increased fraction of precipitation falls as rain rather than being stored as snow at higher elevations and precipitation that does fall as snow melts earlier in the season (Vicuña et al., 2011). In Chile, where most of the annual precipitation falls in winter (Masiokas et al., 2006), this effect is magnified relative to the Argentine catchments. The periodic occurrence of intense, high magnitude winter precipitation events in

Chile (Viale & Norte, 2009) further contribute to increased variability in high flows as more of these events are anticipated to occur as rain.

4.2. Erosion Rates and Sediment Transport Implications

Erosion rates calculated for the baseline period are in general agreement with previously published erosion rates for the region using both river gauge data and ^{10}Be (Carretier et al., 2013; Val et al., 2018). With the exception of the Río Choapa, the catchment with the lowest observed erosion rates, erosion rate estimates from prior studies using gauge data are within error of the range of estimates produced using both observed and simulated streamflow values in this study (Figure 7 and Table S8). Minimal sediment transport from April through October (Figure 3), excluding the high precipitation Río Tinguiririca catchment, is also consistent with previous work investigating sediment fluxes in a high Andean basin (Mao & Carrillo, 2017).

Currently, sediment fluxes in high Andean basins are tied principally to snow and glacier melt (Mao & Carrillo, 2017), and peaks in average monthly streamflow coincide with peaks in average monthly sediment flux for a majority of the studied catchments during the baseline period (Figure 3). Snowmelt is the dominant source of sediment transfer as temperatures warm through the spring, with glaciers taking on a gradually more important role as air temperatures reach their zenith in the summer (Mao & Carrillo, 2017). At least a subset of catchments, including the Río Choapa and the Río Mendoza, do show signs of sediment supply limitation during the baseline period—a characteristic of many river systems (Topping et al., 2000, 2007). In these catchments, lower average monthly sediment loads are exported during the falling limb of the annual hydrograph than during the rising limb (Figure 3). However, with both snow and glacier coverage expected to decline in the future, researchers predict that both proglacial sources of sediment and hillslope sources delivering sediment via snowmelt will become increasingly disconnected from the fluvial system (Carrillo & Mao, 2020). Under the drier and warmer scenario modeled here, a retreating JJAS 0°C isotherm removes approximately 19%–36% of each catchment area from a region of snow accumulation. Snow can link hillslope sources of sediment to main channels through snow avalanches, intermittent tributaries, and protracted periods of high runoff generation during snowmelt (Beylich & Laute, 2015; Carrillo & Mao, 2020). In the absence of these linkages, Carrillo and Mao (2020) predict that future high Andean sediment fluxes will become transport-limited instead of supply limited. In high elevation areas where vegetation is already sparse or nonexistent, increased aridity may also result in the downslope migration of stream channel heads and decreased drainage density, further disconnecting sediment supply from the fluvial system (Tucker & Bras, 1998). Predicted decreases in streamflow and stream power (Figure 5a and Table 2) further suggest that within the fluvial system, the potential for geomorphic work and channel incision could be reduced as the climate becomes more arid (Molnar et al., 2006).

If the high Andes do shift toward a transport-limited environment, future sediment flux in this region could look very different than other higher altitude areas like the Alps or Tibetan Plateau or colder regions like the Arctic, where some studies predict net increases in sediment flux (Costa et al., 2018; Kokelj et al., 2013; Lewis & Lamoureux, 2010; D. Li et al., 2020; Syvitski, 2002). In these regions, increases in sediment flux are often linked with increases in air temperature, forcing increased melt from glaciers, snow, or permafrost and leading to increases in meltwater driven streamflow (Costa et al., 2018; D. Li et al., 2020). In one catchment in the Alps, however, warming temperatures are predicted to lead to a decrease in landslide incidence originating from frost weathering (Hirschberg et al., 2021). The most important factor for the high Andes may be that they are possibly already past “peak water,” so that further warming is not expected to yield greater runoff contributions from glaciers and streamflow is broadly expected to decline (Bozkurt et al., 2018; Ragettli et al., 2016).

A word of caution, of course, is that our analysis does not explicitly consider how hillslope processes might alter sediment supply dynamics in the future. Alterations in these dynamics could propagate into changes in the a and b terms of the sediment rating curves (Asselman, 1999; Yang et al., 2007). For this reason, the results presented herein are more appropriately considered as floors on future sediment flux (Lewis & Lamoureux, 2010). While snowmelt is currently believed to be a key link between hillslope sediment sources and stream channels (Carrillo & Mao, 2020), a greater percentage of high precipitation events occurring as rain on exposed slopes could lead to rapid sediment mobilization (Iida et al., 2012). This effect could be pronounced in the Chilean high Andes due to especially intense winter precipitation events (Viale &

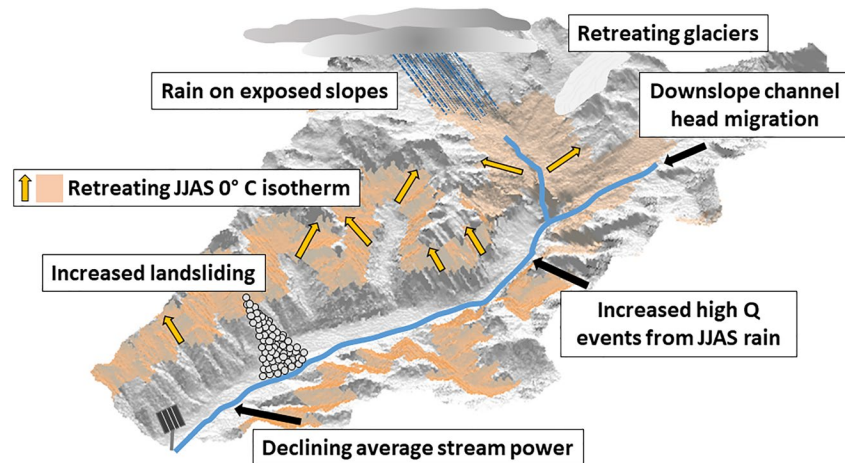


Figure 8. Conceptual diagram showing landscape changes under +4°C of warming. A retreating June, July, August, and September (JJAS) 0°C isotherm (orange arrows) leads to an increase in precipitation events occurring as rain rather than snow on slopes without snow or ice cover. Large wintertime precipitation events on exposed slopes may lead to an increased incidence of landsliding. Reductions in glacier coverage and net annual precipitation are linked with declining stream power and downslope channel head migration.

Norte, 2009) even if total annual precipitation is reduced. In addition, the average slope angle of the regions where JJAS snow cover may disappear exceeds 25° for most catchments, placing these areas at a higher risk of landslide incidence. While landslides are currently more likely to occur during summer storms in the high Andes (Morieras, 2005; Sepúlveda et al., 2015), a higher 0°C isotherm will almost certainly make landsliding a greater risk during a longer portion of the year. Changes in permafrost coverage could also influence both hillslope and streamflow dynamics, and prior studies indicate that conditions are favorable for the presence of permafrost at upper elevations in the high Andes, especially above 4,000 m (Bodin et al., 2010; Pereira et al., 2021). Warming could subject these upslope regions to permafrost melt and consequent slope instability (Baldis & Liaudat, 2019). Figure 8 summarizes many of these possible changes. Quantification of how they may alter sediment supply dynamics in the future warrants further investigation.

In contrast to the two scenarios featuring reductions in precipitation, our warmer scenario shows that rising temperatures alone could produce highly variable future changes to sediment flux, even without explicitly accounting for changes in hillslope supply dynamics. This is especially true for Chile, where the median modeled erosion rates are predicted to increase for a subset of catchments. Increased sediment flux magnitude and variability for Chilean catchments are driven by an increase in the magnitude of outlier, high flow events (Figure 5), such as those associated with intense winter storms originating from the Pacific (Viale & Norte, 2009). In the absence of warming, the impacts of these large winter events, though significant, are nonetheless buffered by precipitation falling as snow at upper elevations, where total precipitation generally increases due to orographic effects. Conversion of some fraction of this high elevation precipitation to rain could contribute to more wintertime flooding in the Chilean catchments (Vicuña et al., 2013). Even if total annual streamflow decreases, the magnitude of these outlier events can accomplish substantial geomorphic work and mobilize significant volumes of sediment (Leopold et al., 1964; Molnar et al., 2006). A greater proportion of winter precipitation falling as rain instead of snow could also increase the frequency of high flow events, since snow is no longer stored to melt out gradually in the spring. Because high rainfall events are associated with regional landslide incidence (Morieras, 2005; Sepúlveda et al., 2015), this is a potential geohazard risk warranting increased study. These outlier events do not appear to have the same impact on the Argentine catchments, where spillover from storm events originating from the west often does not reach the lowlands and is largely concentrated in uninhabited upslope areas (Masiokas et al., 2006). However, even restricted to higher elevations, these events could still lead to short duration, high magnitude streamflow events and may be underestimated in the models. Changes in hillslope processes, such as permafrost melt, could enhance any of these outcomes and should be a focus of future research efforts in the region.

5. Conclusion

We investigated endmember predictions of streamflow and fluvial sediment flux to future changes in air temperature and precipitation for eight catchments in the high Andes of Argentina and Chile. Considering only reductions in precipitation or only increases in air temperature yielded different catchment responses. Future declines in precipitation produced corresponding decreases in streamflow and sediment flux, suggesting that sediment mobilization could become transport-limited, in contrast to other high mountain regions around the world. Considering only warming yielded differing responses in Argentina and Chile. Future warming conditions are expected to produce earlier streamflow arrival and to even produce larger and more frequent high discharge events in Chilean catchments. These characteristics bode for greater future variability in sediment flux in Chile. Importantly, changes in precipitation and temperature considered in this study could also alter sediment supply dynamics in ways not explicitly modeled through the rating curve method employed herein. Supply increases relating to landslide incidence and permafrost melt are two areas in particular warranting future study. Nonetheless, these results indicate that the contrasting hydroclimatic characteristics on either side of the Andes, such as the prevalence of strong winter precipitation events originating from the Pacific, can lead to very different changes in fluvial transport capacity under different climate scenarios. As such, different mitigation strategies may need to be employed on either side of the range to protect water resources and prepare for a changing future.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All data used in this study are from publicly available sources. The HBV model framework used in this study is freely available for download from <https://www.geo.uzh.ch/en/units/h2k/Services/HBV-Model/HBV-Download.html>. The WeaGETS stochastic weather generator is freely available for download from the MATLAB Central File Exchange at <https://www.mathworks.com/matlabcentral/fileexchange/29136-stochastic-weather-generator-weagets>. Streamflow, precipitation, and air temperature data for the Chilean catchments may be obtained via the CAMELS_CL database at <https://doi.pangaea.de/10.1594/PAN-GAEA.894885>. Sediment data for the Chilean catchments are available from the Dirección General de Aguas de Chile at <https://snia.mop.gob.cl/BNAConsultas/reportes>. Streamflow, air temperature, and sediment data for the Argentina catchments are available from the Base de Datos Hidrológica Integrada at <http://bdhi.hidricosargentina.gob.ar>. Precipitation data for the Argentina catchments are available from the CHIRPS database at <https://www.chc.ucsb.edu/data/chirps>. Glacier coverage is available from the Randolph Glacier Inventory at <http://www.glims.org/RGI/randolph60.html>. All future simulations of streamflow are available via CUAHSI Hydroshare at <http://www.hydroshare.org/resource/66371c5590704176979a253ced68bd14>.

Acknowledgments

This work was supported by the National Science Foundation under Grant DGE-1449617 and by Syracuse University's Education Model Program on Water-Energy Research (EMPOWER) and Research Excellence Doctoral Funding (REDF) programs. The authors are grateful for the assistance of Daniel Cielak and Argentina's Dirección de Sistemas de Monitoreo de los Recursos Hídricos and Dirección Nacional de Política Hídrica y Coordinación Federal for providing information on the sampling protocols for sediment data used in this study.

References

- Alvarez-Garretón, C., Mendoza, P. A., Boisier, J. P., Addor, N., Galleguillos, M., Zambrano-Bigiarini, M., et al. (2018). The CAMELS_CL dataset: Catchment attributes and meteorology for large sample studies – Chile dataset. *Hydrology and Earth System Sciences*, 22(11), 5817–5846. <https://doi.org/10.5194/hess-22-5817-2018>
- Asselman, N. E. M. (1999). Suspended sediment dynamics in a large drainage basin: The River Rhine. *Hydrological Processes*, 13, 1437–1450. [https://doi.org/10.1002/\(sici\)1099-1085\(199907\)13:10<1437::aid-hyp821>3.0.co;2-j](https://doi.org/10.1002/(sici)1099-1085(199907)13:10<1437::aid-hyp821>3.0.co;2-j)
- Ayala, Á., Fariás-Barahona, D., Huss, M., Pellicciotti, F., McPhee, J., & Farinotti, D. (2020). Glacier runoff variations since 1955 in the Maipo River basin, in the semiarid Andes of central Chile. *The Cryosphere*, 14, 2005–2027. <https://doi.org/10.5194/tc-14-2005-2020>
- Bagnold, R. (1966). *An approach to the sediment transport problem from general physics*. Professional Paper 422-1. United States Geological Survey. <https://doi.org/10.3133/pp4221>
- Baldis, C. T., & Liaudat, D. T. (2019). Rockslides and rock avalanches in the Central Andes of Argentina and their possible association with permafrost degradation. *Permafrost and Periglacial Processes*, 30(4), 330–347. <https://doi.org/10.1002/ppp.2024>
- BDHI. (2020). *Base de Datos Hidrológica Integrada*. Retrieved from <http://bdhi.hidricosargentina.gob.ar>
- Bergström, S. (1992). *The HBV model: Its structure and applications* (p. 35). Swedish Meteorological and Hydrological Institute (SMHI).
- Beylich, A. A., & Laute, K. (2015). Sediment sources, spatiotemporal variability and rates of fluvial bedload transport in glacier-connected steep mountain valleys in western Norway (Erdalen and Bødalen drainage basins). *Geomorphology*, 228, 552–567. <https://doi.org/10.1016/j.geomorph.2014.10.018>

- Bilotta, G. S., & Brazier, R. E. (2008). Understanding the influence of suspended solids on water quality and aquatic biota. *Water Research*, 42(12), 2849–2861. <https://doi.org/10.1016/j.watres.2008.03.018>
- Bodin, X., Rojas, F., & Brenning, A. (2010). Status and evolution of the cryosphere in the Andes of Santiago (Chile, 33.5°S). *Geomorphology*, 118, 453–464. <https://doi.org/10.1016/j.geomorph.2010.02.016>
- Bozkurt, D., Rojas, M., Boisier, J. P., & Valdivieso, J. (2018). Projected hydroclimate changes over Andean basins in central Chile from downscaled CMIP5 models under the low and high emission scenarios. *Climatic Change*, 150, 131–147. <https://doi.org/10.1007/s10584-018-2246-7>
- Buchhorn, M., Smets, B., Bertels, L., Lesiv, M., Tsendbazar, N.-E., Herold, M., et al. (2020). *Copernicus global land service: Land cover 100m: Collection 3: Epoch 2019, Globe (Version V3.0.1) [Data set]*. Zenodo. <https://doi.org/10.5281/zenodo.3939050>
- Carretier, S., Regard, V., Vassallo, R., Aguilar, G., Martinod, J., Riquelme, R., et al. (2013). Slope and climate variability control of erosion in the Andes of central Chile. *Geology*, 41(2), 195–198. <https://doi.org/10.1130/G33735.1>
- Carrillo, R., & Mao, L. (2020). Coupling sediment transport dynamics with sediment and discharge sources in a glacial Andean basin. *Water*, 12(12), 3452. <https://doi.org/10.3390/w12123452>
- Chen, J., Brissette, F. P., & Leconte, R. (2012). WeaGETS – A Matlab-based daily scale weather generator for generating precipitation and temperature. *Procedia Environmental Sciences*, 13, 2222–2235. <https://doi.org/10.1016/j.proenv.2012.01.211>
- Cortés, G., Vargas, X., & McPhee, J. (2011). Climatic sensitivity of streamflow timing in the extratropical western Andes Cordillera. *Journal of Hydrology*, 405(1–2), 93–109. <https://doi.org/10.1016/j.jhydrol.2011.05.013>
- Costa, A., Molnar, P., Stutenbecker, L., Bakker, M., Silva, T. A., Schlunegger, F., et al. (2018). Temperature signal in suspended sediment export from an Alpine catchment. *Hydrology and Earth System Sciences*, 22, 509–528. <https://doi.org/10.5194/hess-22-509-2018>
- Court, A. (1962). Measures of streamflow timing. *Journal of Geophysical Research*, 67(11), 4335–4339. <https://doi.org/10.1029/JZ067i011p04335>
- Crespo, S. A., Fernandez, F., Cara, L., Klarian, S., & Laverne, C. (2020). First snow, glacier and groundwater contribution quantification in the upper Mendoza River basin using stable water isotopes. *Isotopes in Environmental and Health Studies*, 56(5–6), 566–585. <https://doi.org/10.1080/10256016.2020.1797713>
- Dawson, E. J., & Macklin, M. G. (1998). Speciation of heavy metals on suspended sediment under high flow conditions in the River Aire, West Yorkshire, UK. *Hydrologic Processes*, 12(9), 1483–1494. [https://doi.org/10.1002/\(SICI\)1099-1085\(199807\)12:9<1483::AID-HYP651>3.0.CO;2-W](https://doi.org/10.1002/(SICI)1099-1085(199807)12:9<1483::AID-HYP651>3.0.CO;2-W)
- DGA. (2020). *Información Oficial Hidrometeorológica y de Calidad de Aguas en Línea*. Retrieved from <https://snia.mop.gob.cl/BNAConsultas/reportes>
- Dirección de Sistemas de Monitoreo de los Recursos Hídricos (SMRH). (2021). Muestro de Material en Suspensión.
- Dussaillant, I., Berthier, E., Brun, F., Masiokas, M., Hugonnet, R., Favier, V., et al. (2019). Two decades of glacier mass loss along the Andes. *Nature Geoscience*, 12, 802–808. <https://doi.org/10.1038/s41561-019-0432-5>
- East, A. E., & Sankey, J. B. (2020). Geomorphic and sedimentary effects of modern climate change: Current and anticipated future conditions in the western United States. *Reviews of Geophysics*, 58, e2019RG000692. <https://doi.org/10.1029/2019RG000692>
- Ferguson, R. I. (1986). River loads underestimated by sediment rating curves. *Water Resources Research*, 22(1), 74–76. <https://doi.org/10.1029/WR022i001p00074>
- Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., et al. (2015). The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Scientific Data*, 2, 150066. <https://doi.org/10.1038/sdata.2015.66>
- Gao, P. (2008). Understanding watershed suspended sediment transport. *Progress in Physical Geography: Earth and Environment*, 32(3), 243–263. <https://doi.org/10.1177/0309133308094849>
- Garreaud, R. D., Alvarez-Garreton, C., Barichivich, J., Boisier, J. P., Christie, D., Galleguillos, M., et al. (2017). The 2010–2015 Megadrought in central Chile: Impacts on regional hydroclimate and vegetation. *Hydrology and Earth System Sciences*, 21(12), 6307–6327. <https://doi.org/10.5194/hess-21-6307-2017>
- Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrologic modelling. *Journal of Hydrology*, 377(1–2), 80–91. <https://doi.org/10.1016/j.jhydrol.2009.08.003>
- Hirschberg, J., Faticchi, S., Bennett, G. L., McArdeil, B. W., Peleg, N., Lane, S. N., et al. (2021). Climate change impacts on sediment yield and debris-flow activity in an alpine catchment. *Journal of Geophysical Research: Earth Surface*, 126, e2020JF005739. <https://doi.org/10.1029/2020JF005739>
- Ibañez, M., Gironás, J., Oberli, C., Chadwick, C., & Garreaud, R. (2020). Daily and seasonal variation of the surface temperature lapse rate and 0°C isotherm height in the western subtropical Andes. *International Journal of Climatology*, 41(S1), E980–E999. <https://doi.org/10.1002/joc.6743>
- Iida, T., Kajihara, A., Okubo, H., & Okajima, K. (2012). Effect of seasonal snow cover on suspended sediment runoff in a mountainous catchment. *Journal of Hydrology*, 428–429, 116–128. <https://doi.org/10.1016/j.jhydrol.2012.01.029>
- Jarvis, A., Reuter, H., Nelson, A., & Guevara, E. (2008). *Hole-filled seamless SRTM data V4*. International Centre for Tropical Agriculture (CIAT). Retrieved from <https://cgarsci.community/data/srtm-90m-digital-elevation-database-v4-1/>
- Kokelj, S. V., Lacelle, D., Lantz, T. C., Tunnicliffe, J., Malone, L., Clark, I. D., & Chin, K. S. (2013). Thawing of massive ground ice in mega slumps drives increases in stream sediment and solute flux across a range of watershed scales. *Journal of Geophysical Research: Earth Surface*, 118, 681–692. <https://doi.org/10.1002/jgrf.20063>
- Leopold, L. B., Wolman, M. G., & Miller, J. P. (1964). *Fluvial processes in geomorphology* (p. 522). San Francisco, CA: W.H. Freeman and Company.
- Lewis, T., & Lamoureux, S. F. (2010). Twenty-first century discharge and sediment yield predictions in a small high Arctic watershed. *Global and Planetary Change*, 71, 27–41. <https://doi.org/10.1016/j.gloplacha.2009.12.006>
- Li, D., Li, Z., Zhou, Y., & Lu, X. (2020). Substantial increases in the water and sediment fluxes in the headwater region of the Tibetan Plateau in response to global warming. *Geophysical Research Letters*, 47(11), e2020GL087745. <https://doi.org/10.1029/2020GL087745>
- Li, D., Overeem, I., Kettner, A. J., Zhou, Y., & Lu, X. (2021). Air temperature regulates erodible landscape, water, and sediment fluxes in the permafrost-dominated catchment on the Tibetan Plateau. *Water Resources Research*, 57, e2020WR028193. <https://doi.org/10.1029/2020WR028193>
- Li, L., Ni, J., Chang, F., Yue, Y., Frolova, N., Magritsky, D., et al. (2020). Global trends in water and sediment fluxes of the world's large rivers. *Science Bulletin*, 65(1), 62–69. <https://doi.org/10.1016/j.scib.2019.09.012>
- Mao, L., & Carrillo, R. (2017). Temporal dynamics of suspended sediment transport in a glacierized Andean basin. *Geomorphology*, 287, 116–125. <https://doi.org/10.1016/j.geomorph.2016.02.003>

- Masiokas, M. H., Villalba, R., Luckman, B. H., Le Quesne, C., & Aravena, J. C. (2006). Snowpack variations in the Central Andes of Argentina and Chile, 1951–2005: Large-scale atmospheric influences and implications for water resources in the region. *Journal of Climate*, 19(24), 6334–6352. <https://doi.org/10.1175/JCLI3969.1>
- McClain, M. E., & Naiman, R. J. (2008). Andean influences on the biogeochemistry and ecology of the Amazon River. *BioScience*, 58(4), 325–338. <https://doi.org/10.1641/B580408>
- Molnar, P., Anderson, R. S., Kier, G., & Rose, J. (2006). Relationships among probability distributions of stream discharges in floods, climate, bed load transport, and river incision. *Journal of Geophysical Research: Earth Surface*, 111, F02001. <https://doi.org/10.1029/2005JF000310>
- Morieras, S. M. (2005). Climatic effect of ENSO associated with landslide occurrence in the Central Andes, Mendoza Province, Argentina. *Landslides*, 2, 53–59. <https://doi.org/10.1007/s10346-005-0046-4>
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I – A discussion of principles. *Journal of Hydrology*, 10(3), 282–290. [https://doi.org/10.1016/0022-1694\(70\)90255-6](https://doi.org/10.1016/0022-1694(70)90255-6)
- Pepin, E., Carretier, S., Guyot, J. L., & Escobar, F. (2010). Specific suspended sediment yields of the Andean rivers of Chile and their relationship to climate, slope and vegetation. *Hydrological Sciences Journal*, 55(7), 1190–1205. <https://doi.org/10.1080/02626667.2010.512868>
- Pereira, S. R., Marquardt, C., Beriain, E., & Lambert, F. (2021). Permafrost evolution in a mountain catchment near Santiago de Chile. *Journal of South American Earth Sciences*, 109, 103293. <https://doi.org/10.1016/j.jsames.2021.103293>
- Ragettli, S., Immerzeel, W. W., & Pellicciotti, F. (2016). Contrasting climate change impact on river flows from high-altitude catchments in the Himalayan and Andes Mountains. *Proceedings of the National Academy of Sciences of the United States of America*, 113(33), 9222–9227. <https://doi.org/10.1073/pnas.1606526113>
- RGI Consortium. (2017). *Randolph Glacier Inventory – A dataset of global glacier outlines: Version 6.0: Technical Report, Global Land Ice Measurements from Space*. Digital Media. <https://doi.org/10.7265/N5-RGI-60>
- Rivera, J. A., Marianetti, G., & Hinrichs, S. (2018). Validation of CHIRPS precipitation dataset along the Central Andes of Argentina. *Atmospheric Research*, 213, 437–449. <https://doi.org/10.1016/j.atmosres.2018.06.023>
- Rivera, J. A., Penalba, O. C., Villalba, R., & Araneo, D. C. (2017). Spatio-temporal patterns of the 2010–2015 extreme hydrological drought across the Central Andes, Argentina. *Water*, 9(9), 652. <https://doi.org/10.3390/w9090652>
- Runkel, R. L., Crawford, C. G., & Cohen, T. A. (2004). *Load estimator (LOADEST): A FORTRAN program for estimating constituent loads in streams and rivers* (p. 75). United States Geological Survey Techniques and Methods Book 4, Chapter A5. United States Geological Survey. <https://doi.org/10.3133/tm4A5>
- Rusticucci, M., Zazulie, N., & Raga, G. B. (2014). Regional winter climate of the southern central Andes: Assessing the performance of ERA-Interim for climate studies. *Journal of Geophysical Research: Atmospheres*, 119, 8568–8582. <https://doi.org/10.1002/2013JD021167>
- Seibert, J., & Vis, M. J. P. (2012). Teaching hydrological modeling with a user-friendly catchment-runoff-model software package. *Hydrology and Earth System Sciences*, 16, 3315–3325. <https://doi.org/10.5194/hess-16-3315-2012>
- Sepúlveda, S. A., Moreiras, S. M., Lara, M., & Alfaro, A. (2015). Debris flow in the Andean ranges of central Chile and Argentina triggered by 2013 summer storms: Characteristics and consequences. *Landslides*, 12, 115–133. <https://doi.org/10.1007/s10346-014-0539-0>
- Singh, R., Wagener, T., Crane, R., Mann, M. E., & Ning, L. (2014). A vulnerability driven approach to identify adverse climate and land use change combinations for critical hydrologic indicator thresholds: Application to a watershed in Pennsylvania, USA. *Water Resources Research*, 50(4), 3409–3427. <https://doi.org/10.1002/2013WR014988>
- Snyder, N. P., Rubin, D. M., Alpers, C. N., Childs, J. R., Curtis, J. A., Flint, L. E., & Wright, S. A. (2004). Estimating accumulation rates and physical properties of sediment behind a dam: Englebright Lake, Yuba River, northern California. *Water Resources Research*, 40, W11301. <https://doi.org/10.1029/2004WR003279>
- Somers, L. D., McKenzie, J. M., Mark, B. G., Lagos, P., Ng, G.-L. C., Wickert, A. D., et al. (2019). Groundwater buffers decreasing glacier melt in an Andean watershed – But not forever. *Geophysical Research Letters*, 46(22), 13016–13026. <https://doi.org/10.1029/2019GL084730>
- Syvitski, J. P. M. (2002). Sediment discharge variability in Arctic rivers: Implications for a warmer future. *Polar Research*, 21(2), 323–330. <https://doi.org/10.3402/polar.v21i2.6494>
- Thomas, B. F., & Famiglietti, J. S. (2019). Identifying climate-induced groundwater depletion in GRACE observations. *Scientific Reports*, 9, 4124. <https://doi.org/10.1038/s41598-019-40155-y>
- Thorntwaite, C. W., & Mather, J. R. (1955). *The water balance* (Vol. 8, p. 101). Drexel Institute of Climatology, Laboratory of Climatology: Publications in Climatology.
- Topping, D. J., Rubin, D. M., & Melis, T. S. (2007). Coupled changes in sand grain size and sand transport driven by changes in the upstream supply of sand in the Colorado River: Relative importance of changes in bed-sand grain size and bed-sand area. *Sedimentary Geology*, 202(3), 538–561. <https://doi.org/10.1016/j.sedgeo.2007.03.016>
- Topping, D. J., Rubin, D. M., & Vierra, L. E. (2000). Colorado River sediment transport: 1. Natural sediment supply limitation and the influence of the Glen Canyon Dam. *Water Resources Research*, 36(2), 515–542. <https://doi.org/10.1029/1999WR900285>
- Tucker, G. E., & Bras, R. L. (1998). Hillslope processes, drainage density, and landscape morphology. *Water Resources Research*, 34(10), 2751–2764. <https://doi.org/10.1029/98WR01474>
- Val, P., Venerdini, A. L., Ouimet, W., Alvaredo, P., & Hoke, G. D. (2018). Tectonic control of erosion in the southern Central Andes. *Earth and Planetary Science Letters*, 482, 160–170. <https://doi.org/10.1016/j.epsl.2017.11.004>
- Viale, M., & Garreaud, R. (2015). Orographic effects of the subtropical and extratropical Andes on upwind precipitating clouds. *Journal of Geophysical Research: Atmospheres*, 120(10), 4962–4974. <https://doi.org/10.1002/2014jd023014>
- Viale, M., & Norte, F. A. (2009). Strong cross-barrier flow under stable conditions producing intense winter orographic precipitation: A case study over the subtropical Central Andes. *Weather and Forecasting*, 24(4), 1009–1031. <https://doi.org/10.1175/2009WAF2222168.1>
- Viale, M., & Nuñez, M. N. (2011). Climatology of winter orographic precipitation over the subtropical Central Andes and associated synoptic and regional characteristics. *Journal of Hydrometeorology*, 12(4), 481–507. <https://doi.org/10.1175/2010JHM1284.1>
- Vicuña, S., Garreaud, R. D., & McPhee, J. (2011). Climate change impacts on the hydrology of a snowmelt driven basin in semi-arid Chile. *Climatic Change*, 105, 469–488. <https://doi.org/10.1007/s10584-010-9888-4>
- Vicuña, S., Gironás, J., Meza, F. J., Cruzat, M. L., Jelinek, M., Bustos, E., et al. (2013). Exploring possible connections between hydrological extreme events and climate change in central south Chile. *Hydrologic Sciences Journal*, 58(8), 1598–1619. <https://doi.org/10.1080/02626667.2013.840380>
- Viviroli, D., Archer, D. R., Buytaert, W., Fowler, H. J., Greenwood, G. B., Hamlet, A. F., et al. (2011). Climate change and mountain water resources: Overview and recommendations for research, management and policy. *Hydrology and Earth System Sciences*, 15, 471–504. <https://doi.org/10.5194/hess-15-471-2011>

- Wulf, H., Bookhagen, B., & Scherler, D. (2012). Climatic and geologic controls on suspended sediment flux in the Sutlej River Valley, western Himalaya. *Hydrology and Earth System Sciences*, 16, 2193–2217. <https://doi.org/10.5194/hess-16-2193-2012>
- Yang, G., Chen, Z., Yu, F., Wang, Z., Zhao, Y., & Wang, Z. (2007). Sediment rating parameters and their implications: Yangtze River, China. *Geomorphology*, 85, 166–175. <https://doi.org/10.1016/j.geomorph.2006.03.016>